(3) $\left(D^2+2D+3\right)y=0$

(4) $\frac{d^2y}{dx^2}-2k\frac{dy}{dx}+k^2y=0$

(5) $\left(D^2-2D-15\right)y=0$ given that $\frac{dy}{dx}=0$ and $\frac{d^2y}{dx^2}=2$ when $x=0$

(6) $\left(4D^2+4D-3\right)y=e^{2x}$

(7) $\frac{d^2y}{dx^2}+16y=0$

(8) $\left(D^2-3D+2\right)y=e^{3x}$ which shall vanish for $x=0$ and for $x=\log 2$

(9) $\left(D^2+D-6\right)y=e^{3x}+e^{-3x}$

(10) $\left(D^2-10D+25\right)y=4e^{5x}+5$

(11) $\left(4D^2+16D+15\right)y=4e^{\frac{-3}{2}x}$

(12) $\left(3D^2+D-14\right)y=13e^{2x}$

(13) Suppose that the quantity demanded $Q_d=13-6p+2\frac{dp}{dt}+\frac{d^2p}{dt^2}$ and quantity supplied $Q_s=-3+2p$ where p is the price. Find the equilibrium price for market clearance.

Exercise 4.6

Choose the Correct answer

1. The degree of the differential equation $\frac{d^4y}{dx^4}-\left(\frac{d^2y}{dx^2}\right)^4+\frac{dy}{dx}=3$

(a) 1 (b) 2 (c) 3 (d) 4

2. The order and degree of the differential equation $\sqrt{\frac{d^2y}{dx^2}}=\sqrt{\frac{dy}{dx}+5}$ are respectively

(a) 2 and 3 (b) 3 and 2
(c) 2 and 1 (d) 2 and 2

3. The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^{\frac{3}{2}}-\sqrt{\left(\frac{dy}{dx}\right)}-4=0$ are respectively.

(a) 2 and 6 (b) 3 and 6
(c) 1 and 4 (d) 2 and 4

4. The differential equation $\left(\frac{dx}{dy}\right)^3+2y^{\frac{1}{2}}=x$ is

(a) of order 2 and degree 1
(b) of order 1 and degree 3
(c) of order 1 and degree 6
(d) of order 1 and degree 2

5. The differential equation formed by eliminating a and b from $y=ae^x+be^{-x}$ is

(a) $\frac{d^2y}{dx^2}-y=0$ (b) $\frac{d^2y}{dx^2}-\frac{dy}{dx}=0$
(c) $\frac{d^2y}{dx^2}=0$ (d) $\frac{d^2y}{dx^2}-x=0$

6. If $y=cx+c-c^3$ then its differential equation is

(a) $y=x\frac{dy}{dx}+\frac{dy}{dx}-\left(\frac{dy}{dx}\right)^3$

(b) $y+\left(\frac{dy}{dx}\right)^3=x\frac{dy}{dx}-\frac{dy}{dx}$

(c) $\frac{dy}{dx}+y=\left(\frac{dy}{dx}\right)^3-x\frac{dy}{dx}$

(d) $\frac{d^3y}{dx^3}=0$

7. The integrating factor of the differential equation $\frac{dx}{dy}+Px=Q$ is

(a) $e^{\int Pdx}$ (b) $\int Pdx$
(c) $\int Pdy$ (d) $e^{\int Pdy}$

8. The complementary function of $\left(D^2+4\right)y=e^{2x}$ is

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